LESSON 8: ADVANCED ANALYTICS

Advanced Analytical & Testing Theory

Bridging Data Science and SQL Logic

The Strategic Mindset Shift

Traditional Analysis

Focuses on describing the past. Answers "What happened?" using simple aggregations and basic filters in the database.

Analytical Testing

Focuses on validating business logic. Answers "Why did it change?" and "Is this pattern statistically valid?"

RFM: The Core of Segmentation

Understanding RFM Logic

The **RFM Framework** is a testing method used to group customers based on three variables:

- **Recency:** Time since last purchase.
- **Frequency:** How often they buy.
- **Monetary:** Total revenue generated.

In SQL, we use these to test customer health and predict future churn.

Cohort Analysis Theory

Definition

Grouping users by a shared starting event, usually their first purchase date.

Retention

Measuring how many members of a group remain active over months or years.

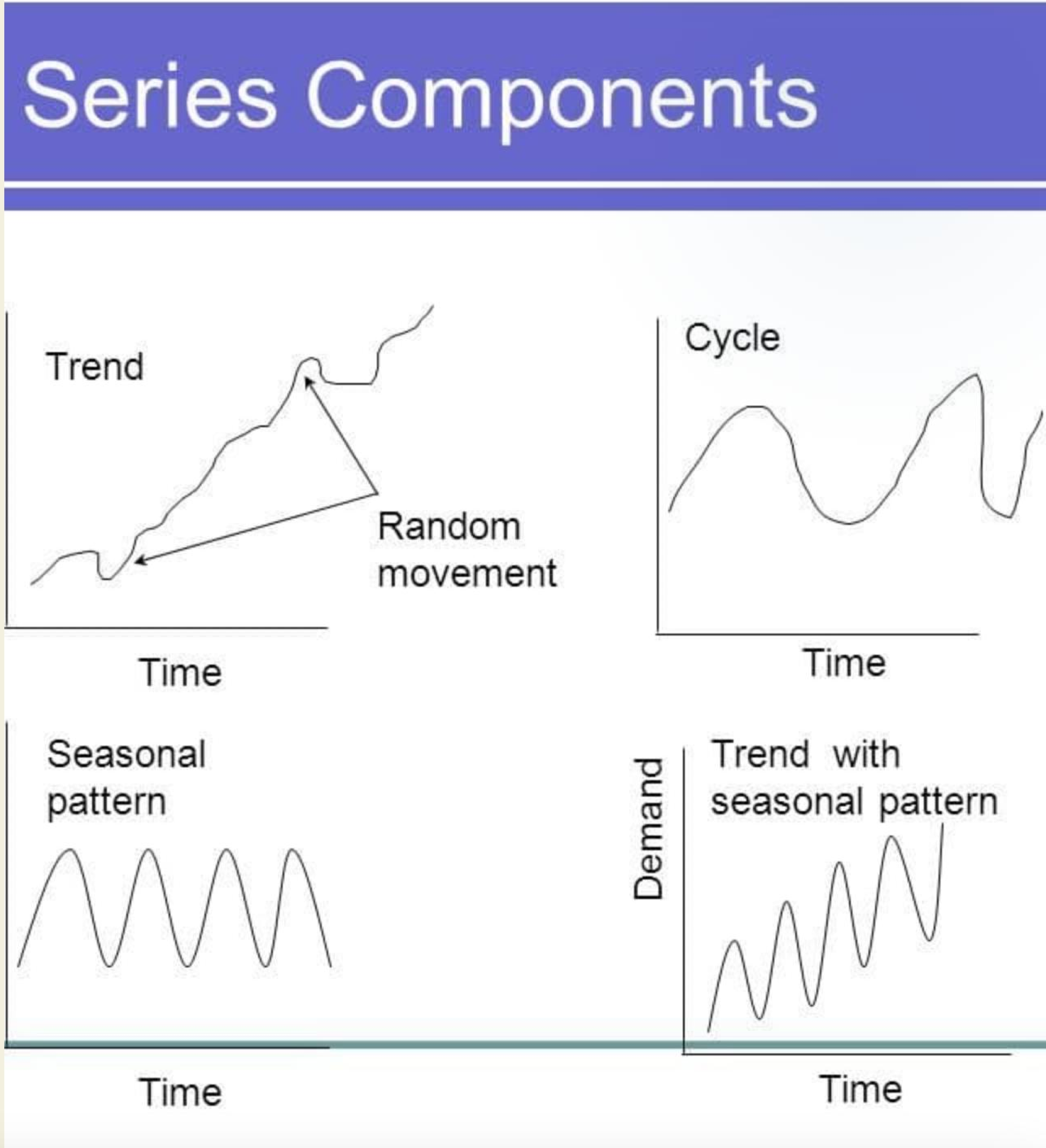
Lifecycle

Identifying the "Golden Period" where customers are most likely to buy again.

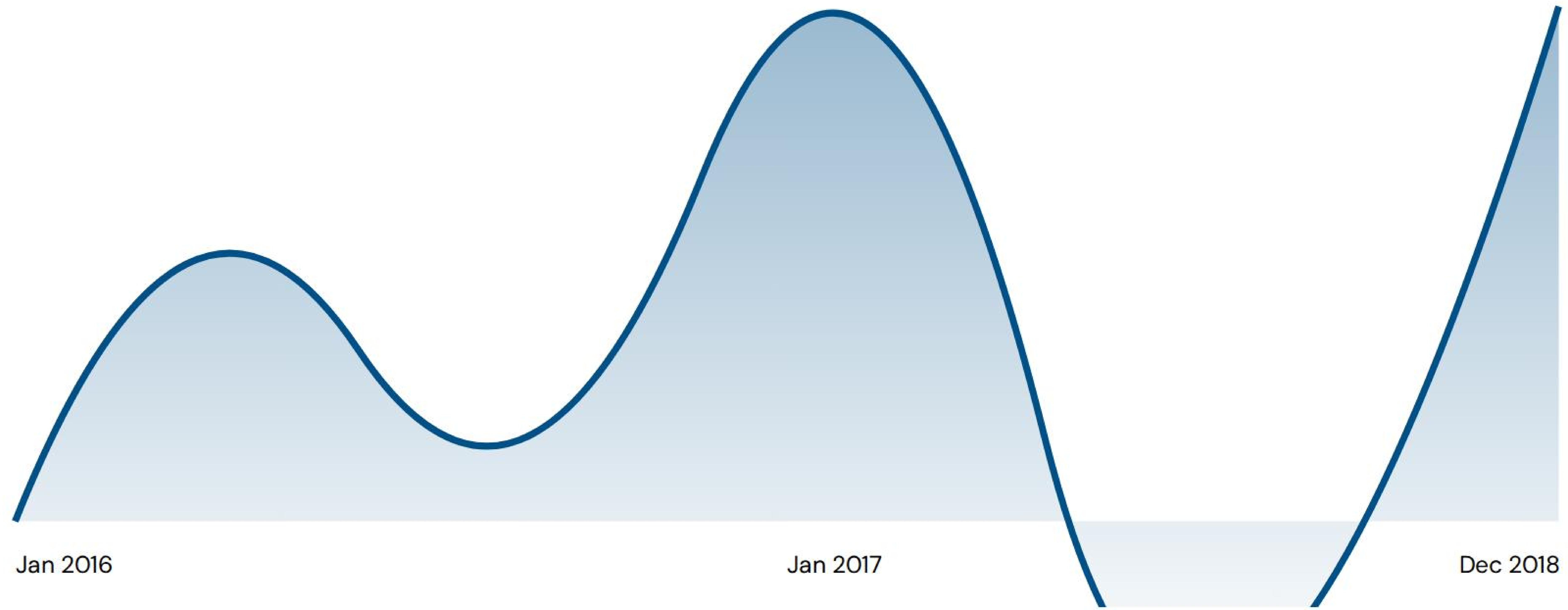
Time Series Dynamics

Trend vs. Seasonality

Testing for **Trend** helps us see if the BikeStore is growing year-over-year.
Testing for **Seasonality** identifies predictable patterns, like bike sales spiking in Summer.
Using **Moving Averages**, we can "smooth" the noise of daily fluctuations to see the truth.



Testing Sales Cycles (2016-2018)



Testing the rolling average allows us to separate "One-time spikes" from "Sustainable growth."

A/B Testing & Impact Analysis

95%

CONFIDENCE INTERVAL

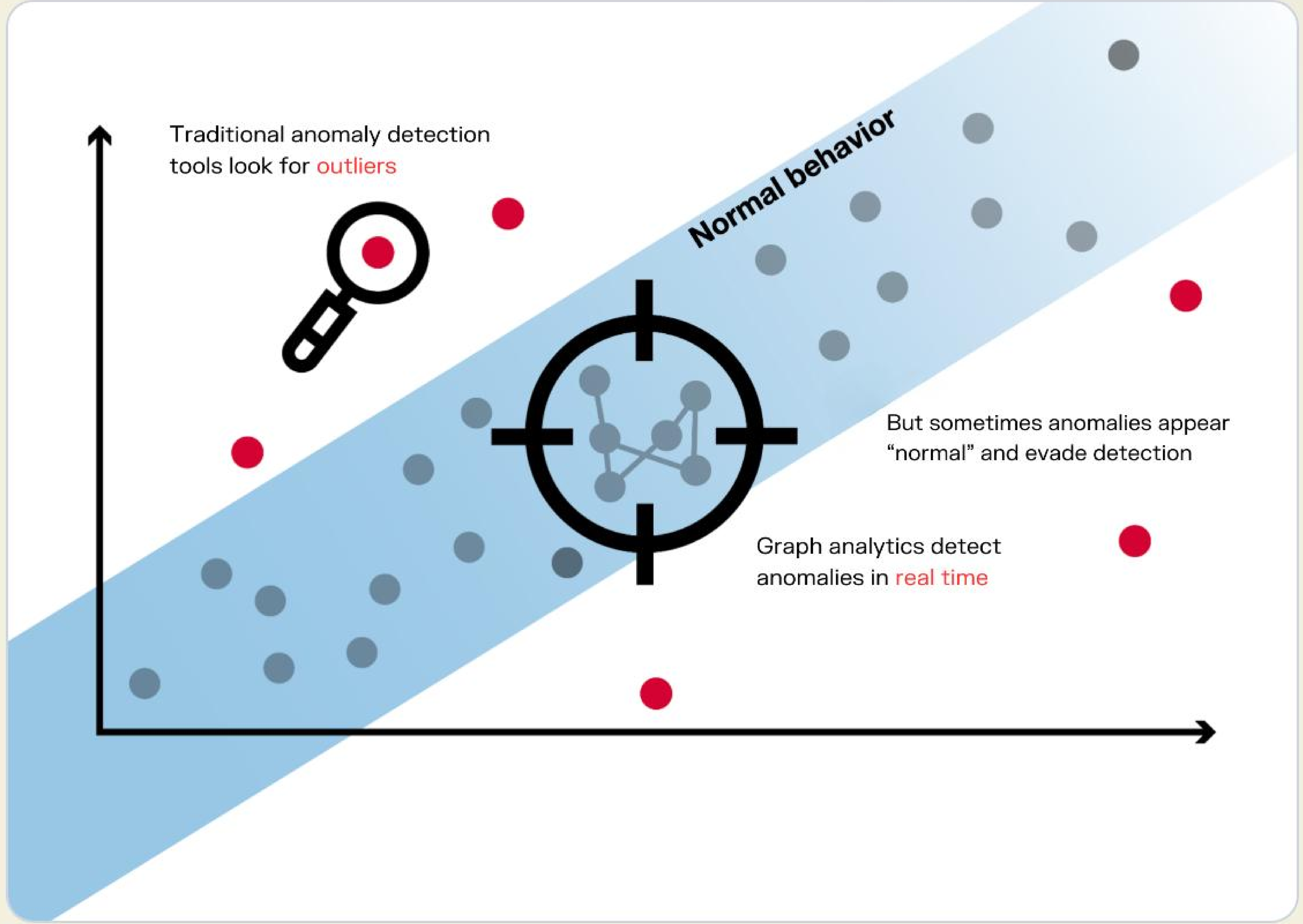
The Scientific Method

A/B testing is not just guessing. It is the practice of comparing a **Control Group** (Standard Prices) against a **Variant Group** (20% Discount).

We use SQL to calculate if the difference in sales is **Statistically Significant** or just luck.

Anomaly Detection Logic

- **Outlier Discovery:** Finding data points that deviate significantly from the norm (e.g., $10,000 order for a $50 bike).
- **Fraud Monitoring:** Using statistical thresholds to flag suspicious activity in the sales.orders table.
- **Data Quality:** Identifying zero-price products or missing order items that break reports.



Mapping Theory to BikeStores SQL

Lesson 8 Topic	SQL Target Table	Business Question (Test Case)
Segmentation	sales.customers	Who are our Top 10% spenders?
Cohort Analysis	sales.orders	How many 2016 customers returned in 2017?
Time Series	sales.orders	What is the monthly sales trend?
A/B Testing	sales.order_items	Did the 20% discount increase volume?
Anomaly Detection	production.stocks	Which products are critically low in stock?

"

"Data is just numbers until you test the logic behind it. Advanced Analysis is the guardrail of business truth."

— Strategic Data Principle

Questions?

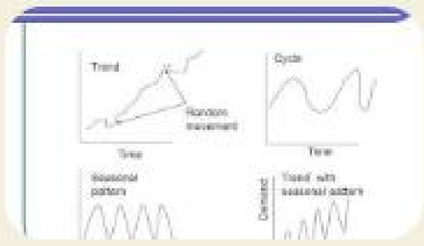
Advanced Analytical Techniques Theory

Up Next: Practical SQL Implementation

Image Sources

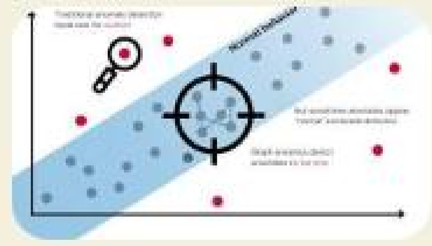
https://businessmodelanalyst.com/wp-content/uploads/2026/01/Ralph-Lauren-Target-Market-1024x576.png

Source: businessmodelanalyst.com



https://fiveable.me/_next/image?url=https%3A%2F%2Fstorage.googleapis.com%2Fstatic.prod.fiveable.me%2Fsearch-images%252F%2522Fundamental_elements_of_time_series_data%253A_trend_seasonality_cycles_examples_like_stock_prices_GDP.%2522-7999.1532720149.jpg&w=3840&q=75

Source: fiveable.me



https://linkurious.com/images/uploads/2024/01/Graph-and-anomaly-detection.png

Source: linkurious.com